

# Phys 115B HW 9

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March 7, 2026

## 1 D

**Question 1.** Consider the parity operator  $\hat{\Pi}$  in three dimensions, such that  $\hat{\Pi}\psi(\mathbf{r}) = \psi(-\mathbf{r})$ .

(a) Show that for states  $\psi$  expressed in polar coordinates, the action of the parity operator can be written as

$$\hat{\Pi}\psi(r, \theta, \phi) = \psi(r, \pi - \theta, \phi + \pi)$$

(b) Show that for hydrogenic energy eigenstates  $\psi_{nlm}(r, \theta, \phi)$  we have

$$\hat{\Pi}\psi_{nlm} = (-1)^l \psi_{nlm}(r, \theta, \phi)$$

*Proof.*

(a) Given the parity operator has the coordinate change  $\mathbf{r} \mapsto -\mathbf{r}$ , one has the following on each component:

$$x : r \sin(\theta) \cos(\phi) \mapsto -r \sin(\theta) \cos(\phi) = r \sin(\pi - \theta) \cos(\phi + \pi) \quad (1)$$

$$y : r \sin(\theta) \sin(\phi) \mapsto -r \sin(\theta) \sin(\phi) = -r \sin(\pi - \theta) \sin(\phi) = r \sin(\theta) \sin(\phi + \pi) \quad (2)$$

$$z : r \cos(\theta) \mapsto -r \cos(\theta) = r \cos(\pi - \theta) \quad (3)$$

So, one transformation on the spherical coordinates that's compatible with the parity operator is  $r \mapsto r$ ,  $\theta \mapsto \pi - \theta$ , and  $\phi \mapsto \phi + \pi$ .

Hence, using  $r, \theta, \phi$  as the input, one has:

$$\hat{\Pi}\psi(r, \theta, \phi) = \psi(r, \pi - \theta, \phi + \pi) \quad (4)$$

(b) Since for  $\psi_{nlm} = R_{n,l}(r)Y_l^m(\theta, \phi)$  in general, then the only one affected by the parity is  $Y_l^m(\theta, \phi)$ .

Recall the formula  $Y_l^m(\theta, \phi) = e^{im\phi} P_l^m(\cos(\theta))$ , where  $P_l^m$  is given as follow:

$$P_l^m(z) = (1 - z^2)^{|m|/2} \left( \frac{d}{dz} \right)^{|m|} P_l(z), \quad P_l(z) := \frac{1}{2^l l!} \left( \frac{d}{dz} \right)^l (z^2 - 1)^l \quad (5)$$

Which, it can be shown inductively, that all  $P_l(z)$  only contains the monomials with the degree of same parity with  $l$ ; then, one can also inductively shown  $\left( \frac{d}{dz} \right)^{|m|} P_l(z)$  has all monomial with degree of same

parity with  $l - |m|$ , showing  $P_l^m(-z) = (-1)^{l-|m|}P_l^m(z)$ . Finally, one has  $e^{im(\phi+\pi)} = (-1)^{|m|}e^{im\phi}$ . Hence, with  $\cos(\pi - \theta) = -\cos(\theta)$ , one has the following:

$$\hat{\Pi}\psi_{nlm} = \psi_{nlm}(r, \pi - \theta, \phi + \pi) = R_{n,l}(r)e^{im(\phi+\pi)}P_l^m(\cos(\pi - \theta)) \quad (6)$$

$$= R_{n,l}(r)(-1)^{|m|}e^{im\phi}P_l^m(-\cos(\theta)) = R_{n,l}(-1)^{|m|}e^{im\phi}(-1)^{l-|m|}P_l^m(\cos(\theta)) \quad (7)$$

$$= (-1)^l R_{n,l}e^{im\phi}P_l^m(\cos(\theta)) = (-1)^l \psi_{nlm}(r, \theta, \phi) \quad (8)$$

□

## 2 D

**Question 2.** We can classify operators depending on how they transform under parity,  $\hat{Q} \rightarrow \hat{\Pi}^\dagger \hat{Q} \hat{\Pi}$  (see example 6.2 or Table 6.1 in the book can be useful).

(a) In one dimension, show that position and momentum are odd under parity.

(b) In three dimensions, operators that are invariant under parity and commute with  $\hat{L}$  are called scalar operators,  $\hat{\Pi}^\dagger \hat{Q}_s \hat{\Pi} = \hat{Q}_s$ . Operators that flip sign under parity and commute with  $\hat{L}$  are called pseudo-scalar operators,  $\hat{\Pi}^\dagger \hat{Q}_s \hat{\Pi} = -\hat{Q}_s$ . Show that scalar operators satisfy  $[\hat{\Pi}, \hat{Q}_s] = 0$ , while pseudoscalar operators satisfy

$$\{\hat{\Pi}, \hat{Q}_s\} = 0$$

(where the curly brackets denote the anti-commutator).

(c) Operators that flip sign under parity and whose commutators with  $\hat{L}$  are like those of  $\hat{L}$  are called vector operators, such that  $\hat{\Pi}^\dagger \hat{Q}_v \hat{\Pi} = -\hat{Q}_v$ . Operators that are invariant under parity and whose commutators with  $\hat{L}$  are like those of  $\hat{L}$  are called pseudovector or axial vector operators,  $\hat{\Pi}^\dagger \hat{Q}_a \hat{\Pi} = \hat{Q}_a$ . Show that vector operators satisfy

$$\{\hat{\Pi}, \hat{Q}_v\} = 0$$

and pseudo-vectors satisfy

$$[\hat{\Pi}, \hat{Q}_a] = 0.$$

(d) Give an example of a scalar operator, pseudoscalar operator, a vector operator and an axial or pseudo vector operator (feel free to Google or use the book).

*Proof.* Here, it uses the fact that  $\hat{\Pi}^\dagger = \hat{\Pi}^{-1} = \hat{\Pi}$ .

(a) Let's first recall some tools, for instance  $\langle x|p\rangle = e^{ipx/\hbar}$ . Then, we have the following:

$$\hat{P}i^\dagger \hat{x} \hat{\Pi} |x\rangle = \hat{\Pi} \hat{x} |-x\rangle = -x \hat{P}i |-x\rangle = -x |x\rangle \quad (9)$$

Since above is true for all  $x \in \mathbb{R}$ , it shows  $\hat{\Pi}^\dagger \hat{x} \hat{\Pi} = -\hat{x}$ .

Similarly, we have the following:

$$\hat{\Pi} |p\rangle = \int \hat{\Pi} |x\rangle \langle x|p\rangle dx = \int e^{ipx/\hbar} |-x\rangle dx \quad (10)$$

$$= \int e^{i(-p)u/\hbar} |u\rangle du = |-p\rangle \quad (11)$$

(Note: Here  $u := -x$ ).

Hence, the full relation on momentum is:

$$\hat{\Pi}^\dagger \hat{p} \hat{\Pi} |p\rangle = \hat{\Pi} \hat{p} |-p\rangle = -p \hat{\Pi} |-p\rangle = -p |p\rangle \quad (12)$$

Since above is true for all  $p \in \mathbb{R}$ , it shows  $\hat{\Pi}^\dagger \hat{p} \hat{\Pi} = -\hat{p}$ .

(b) For scalar operators we have the following:

$$\hat{\Pi}^\dagger \hat{Q}_s \hat{\Pi} = \hat{Q}_s \implies \hat{Q}_s \hat{\Pi} = \hat{\Pi} \hat{Q}_s \implies [\hat{\Pi}, \hat{Q}_s] = 0 \quad (13)$$

For pseudo-scalar operators we have the following:

$$\hat{\Pi}^\dagger \hat{Q}_s \hat{\Pi} = -\hat{Q}_s \implies \hat{Q}_s \hat{\Pi} = -\hat{\Pi} \hat{Q}_s \implies \{\hat{\Pi}, \hat{Q}_s\} = 0 \quad (14)$$

(c) Similarly to the previous part, for vector operators, we have:

$$\hat{\Pi}^\dagger \hat{Q}_v \hat{\Pi} = -\hat{Q}_v \implies \hat{Q}_v \hat{\Pi} = -\hat{\Pi} \hat{Q}_v \implies [\hat{\Pi}, \hat{Q}_v] = 0 \quad (15)$$

For pseudovector operators, we have:

$$\hat{\Pi}^\dagger \hat{Q}_a \hat{\Pi} = \hat{Q}_a \implies \hat{Q}_a \hat{\Pi} = \hat{\Pi} \hat{Q}_a \implies [\hat{\Pi}, \hat{Q}_a] = 0 \quad (16)$$

(d) the free particle hamiltonian  $H = \frac{1}{2m}(p_x^2 + p_y^2 + p_z^2)$  is a scalar operator (since all momentum operators  $p_u$  is odd under parity, so  $\hat{\Pi}^\dagger p_u^2 \hat{\Pi} = (\hat{\Pi}^\dagger p_u \hat{\Pi})(\hat{\Pi}^\dagger p_u \hat{\Pi}) = (-p_u)^2 = p_u^2$ ).

In three dimension, the operator  $x$  is a pseudoscalar operator (since it's proven to have odd parity under 1-dimension, here it's still true).

In three dimension, the operator  $\mathbf{r}$  is a vector operator, since it's odd under parity and its transformation with  $\mathbf{L}$  is some variation of itself.

In three dimension, the operator  $\mathbf{L}$  is a pseudovector operator, since it doesn't change under parity (because for any distinct  $u, v, t \in \{x, y, z\}$ , the component  $L_u = vp_t - tp_v$  up to some sign, which must be even since its summands are products of two odds), while transformation with itself is also a variation of itself.

□

### 3 D

**Question 3.** Consider a particle of mass  $m$  and charge  $q$  confined to a circular ring in the  $x$ - $y$  plane, with radius  $r$ . The system will be driven by an electric field  $E_x$  in the  $x$  direction (which can stimulate photon induced transitions). It is natural to work in polar coordinates for this problem, with  $x = r \cos \theta$ .

- (a) What are the first five energies of the system with  $E_x = 0$ ?
- (b) What are the first five eigenfunctions  $\psi(\theta)$ ? Please normalize them, and pick your basis such that the eigenfunctions are also eigenfunctions of  $L_z$ .
- (c) Explicitly write out a  $3 \times 3$  matrix for the Hamiltonian

$$H' = E_x q x = E_q r \cos \theta$$

This matrix will use the three lowest states above. Extrapolate this result to write a concise selection rule that tells what transitions are allowed for any state.

- (d) What frequencies are absorbed by a “gas” of such rings if they are cold enough to be initially in the ground state?
- (e) What additional frequencies are absorbed by rings initially in the first excited state?

*Proof.* Before starting, given the electric potential  $\varphi$ , one has  $-\nabla\varphi = (E_x, 0)$ , showing that  $\varphi = -E_x x + C$  for some constant  $C \in \mathbb{R}$ . For simplicity, choose  $C = 0$ , hence  $\varphi = E_x x = -E_x r \cos(\theta)$ .

- (a) Suppose  $E_x = 0$ , then the system’s hamiltonian is simply  $H = -\frac{p^2}{2m}$ . By the restriction to  $xy$ -plane,  $p^2 = -\hbar^2 \nabla^2$ , with the restriction that only  $\theta$  varies while  $r$  is fixed, so one has  $\nabla^2 = \frac{1}{r} \frac{\partial}{\partial r} \left( r \frac{\partial}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2}{\partial \theta^2} = \frac{1}{r^2} \frac{\partial^2}{\partial \theta^2}$ .

Hence, any eigenfunction  $\psi$  with energy  $E \geq 0$  satisfies the following equation:

$$E\psi = H\psi = -\frac{\hbar^2}{2mr^2} \frac{\partial^2}{\partial \theta^2} \psi \implies \frac{\partial^2 \psi}{\partial \theta^2} + \left( \frac{r\sqrt{2mE}}{\hbar} \right)^2 \psi = 0 \quad (17)$$

Then, with  $w := \frac{r\sqrt{2mE}}{\hbar}$ , this implies the solutions are  $\psi(\theta) = K e^{iw\theta} + L e^{-iw\theta}$ , with the restriction that it’s a ring (or  $\psi(\theta) = \psi(\theta + 2\pi)$ ), one must have  $w \in \mathbb{Z}$ , showing  $\frac{r\sqrt{2mE}}{\hbar} = n$  for some  $n \in \mathbb{Z}$ , or  $E = \frac{\hbar^2 n^2}{2mr^2}$ .

Hence, the first five lowest energies are:

$$n = 0 \implies E_0 = 0, \quad n = \pm 1 \implies E_1 = \frac{\hbar^2}{2mr^2}, \quad n = \pm 2 \implies E_2 = \frac{2\hbar^2}{mr^2} \quad (18)$$

- (b) Given that the angular momentum  $L_z = -i\hbar \frac{\partial}{\partial \theta}$  (here  $\theta$  is the polar angle), then the eigenfunctions must be of the form  $K e^{a\theta}$  for some  $K, a \in \mathbb{C}$ . Together with the first part, which has the following eigenfunctions:

$$\forall n \in \mathbb{Z}, \quad w = n, \quad \psi_n = K_n e^{in\theta} \quad (19)$$

To normalize it, we need the following condition to hold:

$$1 = \int_0^{2\pi} |\psi_n|^2 d\theta = \int_0^{2\pi} |K_n|^2 d\theta = 2\pi |K_n|^2 \quad (20)$$

This implies that  $|K_n| = \frac{1}{\sqrt{2\pi}}$ . For simplicity, choose  $K_n > 0$ , one has the following eigenfunctions after normalizing:

$$n = 0 \implies \psi_0 = \frac{1}{\sqrt{2\pi}}, \quad n = 1 \implies \psi_1 = \frac{1}{\sqrt{2\pi}} e^{i\theta}, \quad n = -1 \implies \psi_{-1} = \frac{1}{\sqrt{2\pi}} e^{-i\theta} \quad (21)$$

$$n = 2 \implies \psi_2 = \frac{1}{\sqrt{2\pi}} e^{i2\theta}, \quad n = -2 \implies \psi_{-2} = \frac{1}{\sqrt{2\pi}} e^{-i2\theta} \quad (22)$$

- (c) Given the hamiltonian  $H' = Eqr \cos(\theta)$ , with the three lowest energy states  $n = 0, 1, -1$  in order for the basis. Then, the transformation  $\psi \mapsto \cos(\theta)\psi$  can be rewritten as follow for each  $n$ :

$$n = 0, \quad \psi_0 = \frac{1}{\sqrt{2\pi}} \mapsto \frac{1}{\sqrt{2\pi}} \cos(\theta) \quad (23)$$

$$n = 1, \quad \psi_1 = \frac{1}{\sqrt{2\pi}} e^{i\theta} \mapsto \frac{1}{\sqrt{2\pi}} \cos(\theta) e^{i\theta} \quad (24)$$

$$n = -1, \quad \psi_{-1} = \frac{1}{\sqrt{2\pi}} e^{-i\theta} \mapsto \frac{1}{\sqrt{2\pi}} \cos(\theta) e^{-i\theta} \quad (25)$$

Let's do the Fourier Decomposition of the three:

- For  $n = 0$ , based on the fact that  $e^{in\theta} = \cos(n\theta) + i \sin(n\theta)$ , we have the following:

$$\psi_1 + \psi_{-1} = \frac{1}{\sqrt{2\pi}} (e^{i\theta} + e^{-i\theta}) = \frac{1}{\sqrt{2\pi}} 2 \cos(\theta) \implies \frac{1}{\sqrt{2\pi}} \cos(\theta) = \frac{1}{2} \psi_1 + \frac{1}{2} \psi_{-1} \quad (26)$$

- For  $n = 1$ , one requires integral:

$$c_0 = \int_0^{2\pi} \frac{1}{\sqrt{2\pi}} \cos(\theta) e^{i\theta} \frac{1}{\sqrt{2\pi}} d\theta = \frac{1}{2\pi} \left( \int_0^{2\pi} \cos^2(\theta) d\theta + i \int_0^{2\pi} \sin(\theta) \cos(\theta) d\theta \right) \quad (27)$$

$$= \frac{1}{2\pi} \left( \int_0^{2\pi} \frac{1 + \cos(2\theta)}{2} d\theta + \frac{i}{2} \int_0^{2\pi} \sin(2\theta) d\theta \right) \quad (28)$$

$$= \frac{1}{2\pi} \cdot \frac{2\pi}{2} = \frac{1}{2} \quad (29)$$

$$c_1 = \int_0^{2\pi} \frac{1}{\sqrt{2\pi}} \cos(\theta) e^{i\theta} \frac{1}{\sqrt{2\pi}} e^{-i\theta} d\theta = \frac{1}{2\pi} \int_0^{2\pi} \cos(\theta) d\theta = 0 \quad (30)$$

$$c_{-1} = \int_0^{2\pi} \frac{1}{\sqrt{2\pi}} \cos(\theta) e^{i\theta} \frac{1}{\sqrt{2\pi}} e^{i\theta} d\theta = \frac{1}{2\pi} \int_0^{2\pi} \cos(\theta) e^{i2\theta} d\theta \quad (31)$$

$$= \frac{1}{2\pi} \left( \int_0^{2\pi} \cos(\theta) \cos(2\theta) d\theta + i \int_0^{2\pi} \cos(\theta) \sin(2\theta) d\theta \right) \quad (32)$$

$$= 0 \quad (33)$$

- For  $n = -1$ , notice that the imaginary part of  $d_0$  is the negative of  $c_0$  for the case  $n = 1$ , hence  $d_0 = -1$ ; Similarly,  $d_1$  is the complex conjugate of  $c_{-1}$  above, hence  $d_1 = 0$ ; finally,  $d_{-1}$  is the complex conjugate of  $d_1$  above, hence  $d_{-1} = 0$  also.

As a result, in the ordered basis  $\psi_0, \psi_1, \psi_{-1}$ , the transformation  $\psi \mapsto \cos(\theta)\psi$  has the following matrix (when restricting to the first  $3 \times 3$  entries):

$$\begin{pmatrix} 0 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & 0 & 0 \\ \frac{1}{2} & 0 & 0 \end{pmatrix} \quad (34)$$

Then, with  $H' = Eqr \cos(\theta)$ , the above matrix is multiplied by  $Eqr$ , hence we get the following:

$$H' \longrightarrow \frac{Eqr}{2} \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix} \quad (35)$$

Hence, the selection rules states: the state cannot transform to any state with same energy (EX:  $\psi_0$  can't transform to  $\psi_0$ ,  $\psi_{\pm 1}$  can't transform to  $\psi_{\pm 1}$ ). So, the concise form is  $n - m = \pm 1$ , if it transforms from  $n$  to  $m$ .

- (d) Based on the  $H'$ 's matrix, if it's initially in ground state, it can transform from  $\psi_0$  to  $\psi_{\pm 1}$  (from ground state to first excited states), hence the energy it can absorb is  $E_1 - E_0 = \frac{\hbar^2}{2mr^2}$ , so with each photon having energy  $\hbar w$ , the frequency it can absorb is  $w = \frac{E_1 - E_0}{\hbar} = \frac{\hbar}{2mr^2}$ .

- (e) If consider up to  $n = \pm 2$ , notice that one of the first excited state  $\psi_{-1}$  after transformation, say  $\frac{1}{\sqrt{2\pi}} \cos(\theta)e^{-i\theta}$ , its coefficient  $d_{-2}$  (corresponding to  $n = -2$  state  $\psi_{-2} = \frac{1}{\sqrt{2\pi}} e^{-i2\theta}$ ) is given as follow:

$$d_2 = \int_0^{2\pi} \frac{1}{\sqrt{2\pi}} \cos(\theta)e^{-i\theta} \frac{1}{\sqrt{2\pi}} e^{i2\theta} d\theta = \frac{1}{2\pi} \int_0^{2\pi} \cos(\theta)e^{i\theta} d\theta = \frac{1}{2} \quad (36)$$

Hence, it's also possible to jump from energy level  $E_1$  to  $E_2$ . Then, the frequency it can absorb also includes:

$$w' = \frac{E_2 - E_1}{\hbar} = \frac{3\hbar}{2mr^2} \quad (37)$$

□